

Mini Rechargeable Power Supply

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Presented here is a robust, compact and hand-held power supply that delivers 9.2V and 5V up to 1A (approx.) and 3.3V up to 500mA. It also includes a female A-type USB connector so you can power and charge other equipment such as mobile phones. Four white LEDs are included to enable its use as a torch-light as well.

The circuit uses a single rechargeable 3.7V Li-ion battery pack (normally

battery. Even if the battery is fully discharged, it can power the load from external power source without disturbing the output regulation and charging of the battery.

Generally, a power supply derives power from the 230V AC mains and then, using step-down transformer along with circuitry for rectification, filtering and regulation, provides the regulated DC output. The disadvantage is that such power supplies are big and not so portable. The one described



adaptor, USB port or solar panel. This DC input goes to the battery charge management circuit to which a 3.7V single-cell Li-ion battery is connected. It is to be noted here that Li-ion batteries are very sensitive to charging voltage and current, and need a stable charging source. Directly charging Li-ion batteries using any source may reduce the capacity of the battery and degrade it soon.

Battery management. At the heart of the Li-ion battery charge management circuit is the Li-ion battery charger IC BQ24074 from Texas Instruments. It is an integrated Li-ion linear charger and system power path management device for space-limited portable applications. The device operates off either a USB port or a DC adaptor and supports programmable charge currents up to 1.5A. That is, we can charge up to 3000mAh batteries at 0.5C, where C is the battery's capacity.

The BQ24074 features dynamic power path management (DPPM) that powers the system while simultaneously and independently charging the battery. It supports reverse current, short circuit and thermal protection. Its input voltage supports unregulated adaptors.

The battery is charged in three phases: conditioning, constant current and constant voltage. In all these phases, an internal control loop monitors the IC junction temperature and reduces the charge current if the internal temperature threshold is exceeded. All the current settings like charging, input-limiting and termination for the IC are programmable by selecting the right resistor values at appropriate pins.



Fig. 1: Author's prototype

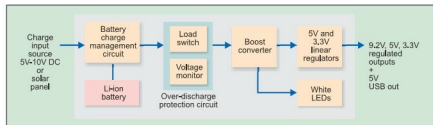


Fig. 2: Block diagram of the mini rechargeable power supply

used in mobile phones and MP3 players). It uses dedicated Li-ion charging IC which monitors undercharge and overcharge conditions to safely charge the Li-ion battery pack. The circuit not only indicates the charging status but also shuts off the load when the battery is discharged to a certain level.

The power supply is so versatile that it accepts input from various power sources like DC wall adaptors, USB or a solar panel to charge the 3.7V

here is a battery-based portable power supply. The author's prototype is shown in Fig. 1.

Circuit and working

The block diagram of the mini rechargeable power supply is shown in Fig. 2 and its circuit diagram in Fig. 3.

Charging input source. The charging input source for the power supply can be anything from 5V to 10V, which can be harnessed using a DC power